IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor:	Jae Jun KO	Examiner:	BOOSALIS, FANI POLYZOS
Serial No.:	16/972,577	Art Unit:	2884
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For: **RADON SENSOR DEVICE USING POLYHEDRAL-SHAPED IONIZATION CHAMBER**

RESPONSE

Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

September 17, 2021

Sir:

In response to the Non-Final Office Action dated August 31, 2021, Applicants provide the following amendments and remarks:

Amendments to the Claims begin on page **2** of this paper.

Remarks begin on page 7 of this paper.

CERTIFICATE OF ELECTRONIC TRANSMISSION
I hereby certify that this document is being electronically transmitted to the Commissioner for Patents via EFS-Web on September 17, 2021.

LUCAS & MERCANTI, LLP
BY: /Yun H. Choe/
Yun H. Choe

AMENDMENTS TO THE CLAIMS

This listing of claims will replace all prior versions, and listings, of claims in the subject application:

Listing of Claims:

1. (Currently Amended) A radon sensor device using a polyhedral-shaped ionization chamber, the radon sensor device comprising:

an ionization chamber having an open side and inner sides surrounded by a first conductor and generating an electrical field therein by applying bias power to the first conductor;

a cover having a first side covered with a second conductor and closing the open side of the ionization chamber such that that first conductor disposed on the inner sides of the ionization chamber and the second conductor are electrically connected;

a probe unit disposed in the ionization chamber and absorbing ion charges produced when alpha (α) decay occurs in the ionization chamber; and

a measurement circuit detecting an alpha particle detection signal by amplifying and processing an electrical micro-signal input from the probe unit into a predetermined magnitude,

wherein the first conductor disposed on the inner sides of the ionization chamber and the second conductor disposed on the first side of the cover are electrically connected through joints between the cover and the ionization chamber.

2. (Original) The radon sensor device of claim 1, wherein the ionization chamber is formed in a hexahedral shape using an insulating material.

3. (Original) The radon sensor device of claim 1, wherein the first conductor disposed on the inner sides of the ionization chamber is formed by coating conductive paint with a predetermined thickness on the inner sides of the ionization chamber.

4. (Original) The radon sensor device of claim 1, wherein the bias power that is applied to the first conductor disposed on the inner sides of the ionization chamber is a DC voltage of 50V to 300V.

5. (Cancelled)

6. (Original) The radon sensor device of claim 1, wherein the cover is a printed circuit board (PCB) having a predetermined thickness and the entire of a first side of the printed circuit board is covered with the second conductor.

7. (Original) The radon sensor device of claim 1, wherein a plurality of holes is formed through the cover so that external air can flow into the ionization chamber or air can flow outside.

8. (Original) The radon sensor device of claim 7, wherein when a volume of the ionization chamber is 100cc, a diameter of the holes formed through the cover is 0.2mm to 0.3mm and the number thereof is 6 to 10.

9. (Original) The radon sensor device of claim 1, wherein the probe unit detects ion charges produced by a collision of air and alpha particles produced when alpha (α) decay occurs in the ionization chamber.

10. (Original) The radon sensor device of claim 1, wherein the probe unit is disposed at a center in the ionization chamber and is formed in a T-shape.

11. (Original) The radon sensor device of claim 10, wherein the probe unit is formed in a T-shape by perpendicularly combining first and second probes that have a rod shape or a straight shape having a predetermined length; and

a first end of the first probe is electrically connected with the measurement circuit through the cover and a second end of the first probe is disposed at the center in the ionization chamber, and a center of the second probe is perpendicularly coupled to the second end of the first probe and is disposed at the center in the ionization chamber such that distances between the inner sides of the ionization chamber and a surrounding portion of the second probe are the same.

12. (Original) The radon sensor device of claim 1, wherein the probe unit is disposed at a center

in the ionization chamber and is formed in an outdoor TV antenna shape.

13. (Original) The radon sensor device of claim 12, wherein the probe unit is formed by combining a first probe having a T-shape and second probes having a predetermined pattern,

wherein the first probe has an upper head formed in a rod shape or a straight shape with a predetermined length and disposed at the center in the ionization chamber, has a lower leg formed in a rod shape or a straight shape with a predetermined length and having a first end perpendicularly connected to a center of the upper head to form a T-shape and a second end electrically connected to the measurement circuit through the cover; and

the second probes are formed in a rod shape or a straight shape with a predetermined length, are coupled at centers to the upper head of the first probe, are longitudinally spaced a predetermined distance from each other, and are coupled perpendicular to or at a predetermined angle with respect to a longitudinal direction of the upper head of the first probe to have a predetermined pattern.

14. (Original) The radon sensor device of claim 12, wherein the probe unit is formed by combining a first probe having a T-shape and second probes having a predetermined pattern,

wherein the first probe has an upper head formed in a rod shape or a straight shape with a predetermined length and disposed at the center in the ionization chamber, has a lower leg formed in a rod shape or a straight shape with a predetermined length and having a first end perpendicularly connected to a center of the upper head to form a T-shape and a second end electrically connected to the measurement circuit through the cover; and

the second probes are combined to vertically and horizontally cross each other at a right angle in a lattice pattern and centers thereof are longitudinally coupled to the upper head of the first probe.

15. (Original) The radon sensor device of claim 14, wherein the second probes are coupled to the upper head of the first probe to vertically and horizontally cross each other at a right angle, whereby the lattice pattern thereof is a square lattice pattern or a skew-crossed pattern.

16. (Original) The radon sensor device of claim 1, wherein the probe unit is disposed in the

ionization chamber such that distances between the inner sides of the ionization chamber and a surrounding portion of the second probe are the same.

17. (Original) The radon sensor device of claim 1, wherein the measurement circuit is disposed on a second side of the cover and is mounted to be electrically connected with the second conductor disposed on the first side of the cover through a bias voltage outputter for outputting the bias power.

18. (Original) The radon sensor device of claim 1, wherein the measurement circuit comprises:

a guard ring installed such that the probe unit is disposed therethrough and absorbing and sensing a leakage current generated between the ionization chamber and the probe unit to a ground;

a noise detector connected between the guard ring and the ground, sends a DC-type leakage current generated between the ionization chamber and the probe to the ground, and detects and outputs AC-type noise;

first and second pre-amplifiers respectively connected to another end of the probe and an output end of the noise detector, and amplifying small electrical signals input from the probe and the noise detector into a predetermined magnitude; and

a differential amplifier having a non-inverting terminal (+) and an inverting terminal (-) respectively connected to output ends of the first and second pre-amplifiers, and offsetting noise signals and outputting an alpha particle detection signal by amplifying a voltage difference of electrical signals pre-amplified through the first and second pre-amplifiers.

19. (Original) The radon sensor device of claim 18, wherein the noise detector is configured by electrically connecting at least any one element of a resistance, a condenser, and a diode in series, in parallel, or in combination of series and parallel.

20. (Original) The radon sensor device of claim 1, further comprising:

an electromagnetic wave-protective cap installed on a second side of the cover to surround the entire measurement circuit and is made of a conductive material to prevent signal disturbance of the measurement circuit due to external electromagnetic wave noise,

wherein the electromagnetic wave-protective cap is installed to be electrically connected with a ground of the measurement circuit.

REMARKS

Reconsideration of the present application, as amended, is respectfully requested.

A. STATUS OF THE CLAIMS

Claims 1-20 were pending.

Claim 1 has been amended by incorporating the limitation of allowable dependent claim 5, which has been cancelled without prejudice.

Thus, claims 1-4 and 6-20 are currently pending.

No new matter have been added.

B. 35 U.S.C. §§ 102 and 103

Claim 1, 6-7, 9, 10, 17, 20 were rejected under 35 U.S.C. 102(a)(2) as being anticipated by Kwon et al (US 2020/0278457 Al).

Claim 2 was rejected under 35 U.S.C. 103 as being unpatentable over Kwon et al (US 2020/0278457 Al) in view of Wedding et al (US 7,791,037 Bl).

Claim 3 was rejected under 35 U.S.C. 103 as being unpatentable over Kwon et al (US 2020/0278457 Al) in view of Barstad et al (US 2,842,674)

Claim 4 was rejected under 35 U.S.C. 103 as being unpatentable over Kwon et al (US 2020/0278457 Al) in view of Esin et al (US 8,415,638 B2).

Claims 5, 8, 11-16, 18-19 were objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

As such, independent claim 1 has been amended to incorporate the limitation of allowable dependent claim 5.

Thus, withdrawal of these rejections is respectfully requested.

C. FEES

The response to the office action is being filed within a shortened period of reply. Thus, no further fee is believed due. If, on the other hand, the United States Patent and Trademark Office determines that any fees are due or Applicants have made any overpayment, the Assistant Commissioner is hereby authorized to debit or credit such sum to Deposit Account No. 02-2275.

Pursuant to 37 C.F.R. § 1.136(a)(3), please treat this and any concurrent or future reply in the subject application that requires a petition for an extension of time for its timely submission as incorporating a petition for extension of time for the appropriate length of time. The fee associated therewith is to be charged to Deposit Account No. 02-2275.

D. CONCLUSION

In view of the arguments presented, Applicants respectfully submit that they have addressed by this Submission each and every one of the matters raised by the Examiner. However, Applicants reserve the right to respond to any outstanding issues which have not been addressed in this Submission. Furthermore, Applicants respectfully urge that the Examiner contact the undersigned with any question.

An early and favorable action on the merits is earnestly solicited.

Respectfully submitted,

LUCAS & MERCANTI, LLP

By: /Yun H. Choe/
Yun H. Choe
Registration No. 61,798

LUCAS & MERCANTI, LLP
30 Broad Street, 21st Floor
New York, New York 10004
Phone: 212-661-8000
Fax: 212-661-8002
YHC/ABC